



DEFENSE HUB



Dashboard & Progress

TASK REPOSITORIES



1. STT Model Pool



2. Speech Benchmarking



3. Filler Data Explorer



4 & 5. TTS Showcase



Practice Meta

Supervisor: Senior Researcher

AITU: Defense Portal

Language: English / Kazakh

# Scientific Practice Dashboard

Empirical timeline, core system achievements, and task progress breakdown for Kazakh Speech Technologies.

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TASKS  
COMPLETED

408K

FILLER  
SPEECH  
SAMPLES

0.05

BEST  
MODEL RTF  
(SOYLE)

## Research Task Progression



### Task 1

DONE

#### STT Model Exploration

Cataloged 5 state-of-the-art ASR models supporting Kazakh. Evaluated structural differences, parameter scales, and licenses to choose candidates for local testing.



### Task 2

DONE

#### Local STT Testing & Benchmarking

Implemented a benchmark harness. Calculated RTF

## Abstract & Core Defense Rationale

This defense portal represents the empirical milestones achieved during the scientific practice in **Kazakh Speech Technologies**. The research explores two critical vectors:

- **Colloquial Robustness:** Benchmarking ASR models on *Shala-Kazakh* code-switching dialects, revealing a 15-20% average drop in WER compared to pure literary speech.